

گزارش سوال ۳ عملی - تمرین ۱ - سیستم عامل

برای این سوال از Debian 13 استفاده کردیم. ابتدا با دستورات زیر پیش نیاز های لازم را نصب کردیم:

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y build-essential libssl-dev flex bison libncurses-dev \
libelf-dev bc dwarves wget tar gzip
```

سپس کد منبع هسته را دانلود و decompress کردیم:

```
wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.1.156.tar.xz
tar -xf linux-6.1.156.tar.xz
cd linux-6.1.156
```

فایل پیکربندی اولیه را ایجاد کردیم:

```
bagheri-timarchi@debian:~/Desktop/linux-6.1.156$ cp /boot/config-$(uname -r) .config
```

سپس با اجرای دستور زیر در پنجره باز شده تنظیمات گفته شده را اعمال کردیم:

```
bagheri-timarchi@debian:~/Desktop/linux-6.1.156$ make menuconfig
```

...



همچنین برای اینکه در مرحله بوت کردن هسته جدید با مشکل decompress کردن مازول ها مواجه نشویم،
فشرده‌سازی آن ها را غیر فعال کردیم:

```
bagheri-timarchi@debian:~/Desktop/linux-6.1.156$ grep -E  
"CONFIG_KERNEL_|CONFIG_MODULE_COMPRESS|CONFIG_RD_" .config  
CONFIG_KERNEL_GZIP=y  
# CONFIG_KERNEL_BZIP2 is not set  
# CONFIG_KERNEL_LZMA is not set  
# CONFIG_KERNEL_XZ is not set  
# CONFIG_KERNEL_LZO is not set  
# CONFIG_KERNEL_LZ4 is not set  
# CONFIG_KERNEL_ZSTD is not set  
CONFIG_RD_GZIP=y  
CONFIG_RD_BZIP2=y  
CONFIG_RD_LZMA=y  
# CONFIG_RD_XZ is not set  
CONFIG_RD_LZO=y  
CONFIG_RD_LZ4=y  
CONFIG_RD_ZSTD=y  
CONFIG_MODULE_COMPRESS_NONE=y  
# CONFIG_MODULE_COMPRESS_GZIP is not set  
# CONFIG_MODULE_COMPRESS_XZ is not set  
# CONFIG_MODULE_COMPRESS_ZSTD is not set
```

حذف درایور های غیر ضروری برای افزایش سرعت build:

```
bagheri-timarchi@debian:~/Desktop/linux-6.1.156$ make localmodconfig  
HOSTCC scripts/kconfig/conf.o  
HOSTLD scripts/kconfig/conf  
using config: '.config'  
...  
#  
# configuration written to .config  
#
```

کامپایل هسته و مازول ها:

```
make -j$(nproc)  
sudo make modules_install  
sudo make install
```

هسته کامپایل شده:

```
Warning: os-prober will be executed to detect other bootable partitions.
Its output will be used to detect bootable binaries on them and create new boot entries.
Found Windows Boot Manager on /dev/nvme0n1p1@/EFI/Microsoft/Boot/bootmgfw.efi
Adding boot menu entry for UEFI Firmware Settings ...
done
```

به روز رسانی grub:

```
bagheri-timarchi@debian:~$ sudo update-initramfs -c -k 6.1.156 [sudo]
password for bagheri-timarchi:
update-initramfs: Generating /boot/initrd.img-6.1.156
```

```
bagheri-timarchi@debian:~/Desktop/linux-6.1.156$ sudo update-grub
Generating grub configuration file ... Found background image:
/usr/share/images/desktop-base/desktop-grub.png
Found linux image: /boot/vmlinuz-6.12.48+deb13-amd64
Found initrd image: /boot/initrd.img-6.12.48+deb13-amd64
Found linux image: /boot/vmlinuz-6.12.43+deb13-amd64
Found initrd image: /boot/initrd.img-6.12.43+deb13-amd64
Found linux image: /boot/vmlinuz-6.1.156 Found initrd image:
/boot/initrd.img-6.1.156 Warning: os-prober will be executed to
detect other bootable partitions. Its output will be used to detect
bootable binaries on them and create new boot entries. Found Windows
Boot Manager on /dev/nvme0n1p1@/EFI/Microsoft/Boot/bootmgfw.efi
Adding boot menu entry for UEFI Firmware Settings ...
done
```

سپس:

```
sudo reboot
```

پس از روشن شدن مجدد سیستم به Advanced options for Debian رفته و Linux 6.1.156 را انتخاب کردیم تا

سیستم از هسته جدید استفاده کند. برای تأیید نسخه هسته استفاده شده از دستور زیر استفاده کردیم:

```
bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$ uname -r
6.1.156
```

سپس ماژول خواسته شده را نوشتیم:

```
bagheri-timarchi@debian:~/Desktop/linux-6.1.156$ mkdir mymodule
bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$ nano
mymodule.c bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$
cat mymodule.c  #include <linux/module.h>  #include <linux/kernel.h>
#include <linux/init.h>

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Your Name");
MODULE_DESCRIPTION("A simple test kernel module");

static int __init mymodule_init(void) {    printk(KERN_INFO
"mymodule: loaded successfully!\n");    return 0;  }

static void __exit mymodule_exit(void) {    printk(KERN_INFO
"mymodule: unloaded successfully!\n");  }

module_init(mymodule_init);
module_exit(mymodule_exit);
```

Makefile آن را نوشتیم:

```
bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$ nano Makefile
bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$ cat Makefile
obj-m += mymodule.o

all:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) modules

clean:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) clean
```

```

bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$ make
make -C /lib/modules/6.1.156/build M=/home/bagheri-
timarchi/Desktop/linux-6.1.156/mymodule modules
CC [M] /home/bagheri-timarchi/Desktop/linux-
6.1.156/mymodule/mymodule.o
MODPOST /home/bagheri-timarchi/Desktop/linux-
6.1.156/mymodule/Module.symvers
CC [M] /home/bagheri-timarchi/Desktop/linux-
6.1.156/mymodule/mymodule.mod.o
LD [M] /home/bagheri-timarchi/Desktop/linux-
6.1.156/mymodule/mymodule.ko
BTF [M] /home/bagheri-timarchi/Desktop/linux-
6.1.156/mymodule/mymodule.ko

```

در قسمت زیر افزودن و سپس حذف موفقیت آمیز ماژول نوشته شده را ببینید:

```

mymodule: bash - Konsole
bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$ sudo insmod mymodule.ko
[sudo] password for bagheri-timarchi:
bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$ sudo rmmod mymodule
bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$ dmesg | tail
dmesg: read kernel buffer failed: Operation not permitted
bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$ sudo dmesg | tail
[ 246.847421] audit: type=1400 audit(1761576488.494:144): apparmor="ALLOWED" operation="open" profile="libreoffice-soffice" name="/home/bagheri-
timarchi/.config/kdedefaults/kcminputrc" pid=3194 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[ 246.847444] audit: type=1400 audit(1761576488.494:145): apparmor="ALLOWED" operation="open" profile="libreoffice-soffice" name="/home/bagheri-
timarchi/.config/kcminputrc" pid=3194 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[ 246.946760] audit: type=1400 audit(1761576488.594:146): apparmor="ALLOWED" operation="open" profile="libreoffice-soffice" name="/usr/share/qt6
/translations/qt_en.qm" pid=3194 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[ 246.956301] audit: type=1400 audit(1761576488.602:147): apparmor="ALLOWED" operation="open" profile="libreoffice-soffice" name="/home/bagheri-
timarchi/.config/kdedefaults/kwinrc" pid=3194 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[ 246.956324] audit: type=1400 audit(1761576488.602:148): apparmor="ALLOWED" operation="open" profile="libreoffice-soffice" name="/home/bagheri-
timarchi/.config/kwinrc" pid=3194 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[ 246.958117] audit: type=1400 audit(1761576488.606:149): apparmor="ALLOWED" operation="open" profile="libreoffice-soffice" name="/home/bagheri-
timarchi/.config/kdedefaults/kwinrc" pid=3194 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[ 570.675959] mymodule: loading out-of-tree module taints kernel.
[ 570.676075] mymodule: module verification failed: signature and/or required key missing - tainting kernel
[ 570.677331] mymodule: loaded successfully!
[ 615.012493] mymodule: unloaded successfully!
bagheri-timarchi@debian:~/Desktop/linux-6.1.156/mymodule$

```